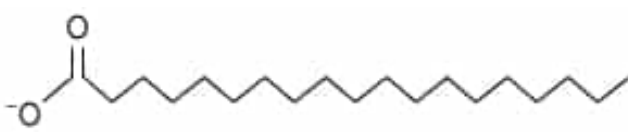
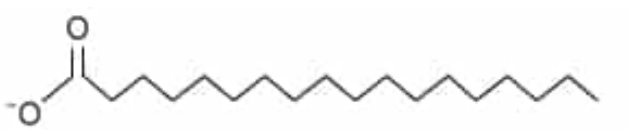
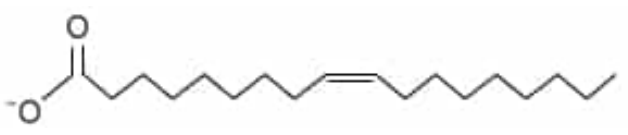
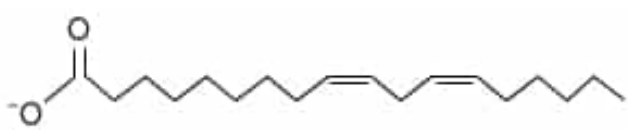


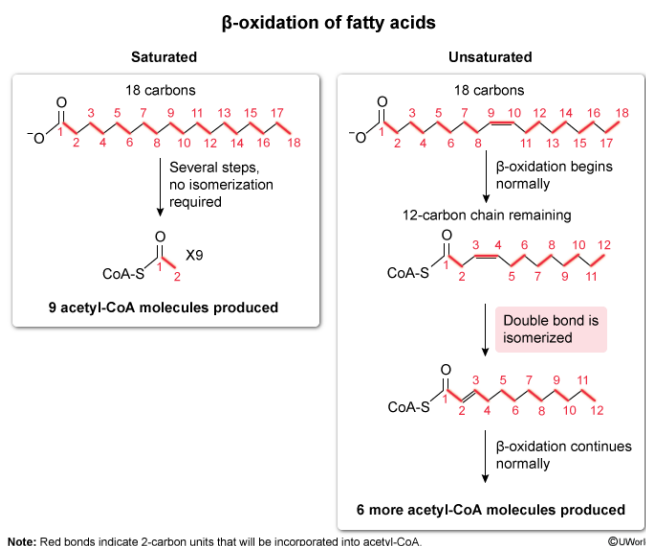
A fatty acid undergoes complete β -oxidation to yield nine molecules of acetyl-CoA. No isomerization reactions occur. The structure of the original fatty acid was:

- ☐ A.  [44%]
- ☒ B.  [45%]
- ☐ C.  [6%]
- ☐ D.  [3%]

Correct

44% answered correctly

Explanation:



Fatty acids are long hydrocarbon chains with a carboxylic acid group on one end. They are degraded through a process called **fatty acid oxidation**, or β -oxidation, which proceeds differently depending on the composition of the fatty acid being oxidized. **All fatty acids** are broken down into multiple **two-carbon** units called **acetyl-CoA**. Fatty acids with an **even number** of carbons yield **half as many** acetyl-CoA units as the **total** number of carbons in the original molecule. If the fatty acid has an **odd number** of carbons, the final **three**

carbons are converted into a compound called **propionyl-CoA**. In other words, for odd-chain fatty acids, the number of acetyl-CoA units produced equals $(\text{total carbons} - 3)/2$.

Unsaturated fatty acids have at least one C=C double bond, typically in the *cis*-conformation. For β -oxidation to proceed, the ***cis*-bonds must be isomerized** to *trans*-bonds, which are introduced as part of the oxidation mechanism. Saturated fatty acids do not have C=C double bonds initially and do not need to undergo isomerization during β -oxidation.

The question states that the β -oxidation resulted in **9 acetyl-CoA units**, so the original fatty acid must have had twice that number of carbons (**18**). The fact that no isomerization reactions were needed indicates that the **fatty acid was saturated**. Only Choice B shows a saturated fatty acid with 18 carbons.

(Choice A) This option shows a saturated fatty acid with 19 carbons. Because it has an odd number of carbons, it will produce 8 acetyl-CoA units (from 16 carbons) and have 1 three-carbon unit (called propionyl-CoA) remaining.

(Choices C and D) These options show unsaturated fatty acids, which require isomerization of the *cis*-double bond.

Educational objective:

Fatty acids can be oxidized to form acetyl-CoA, and those with an even number of carbons produce half as many acetyl-CoA molecules as they have carbons. Fatty acids with an odd number of carbons produce propionyl-CoA in addition to acetyl-CoA. Unsaturated fatty acids require isomerization reactions to convert *cis*-bonds to *trans*-bonds whereas saturated fatty acids do not.

Time spent:0 sec

Subject:
Biochemistry

QID:402025

System:
Metabolic Reactions